

SEMIOTAX: Bridging Human Language and Artificial Intelligence

A) What is it about?

1) Semiotax is an innovative linguistic framework that transforms human sentence structures into precise mathematical equations designed for Natural Language Processing (NLP) and Artificial Intelligence. Historically, computational systems have struggled to capture the deeper, cross-linguistic meaning of human speech because traditional grammar rules vary too wildly across different languages. To resolve this problem, this research proposes that every sentence, regardless of the language it is spoken or written in, shares a universal and mathematically consistent core.

2) The core of this study demonstrates that every sentence operates as an essential ternary structure. This means it is built from three indispensable components: a Referential element (the starting point), a Vectorial connector (the action or verb), and a Terminal element (the endpoint). Using detailed formulas, the author shows how these core components expand internally into binary networks.

3) The most disruptive contribution of this work is the mathematical formulation of "computational thought". This concept bridges neurolinguistics and software engineering by modeling human cognitive processes through unified algorithmic logic. Practically, this top-down methodological approach paves the way for AI that can interpret, structure, and explain human thought with mathematical precision.

B) Relevance

1. What Makes This Work Unique?

1.a) *A Mathematical Interface for Meaning*: Unlike traditional syntax trees or statistical representations, this work introduces a novel framework—Sentence Semiotax—that maps language directly to mathematical equations.

1.b) *Universal Algebraic Invariance*: It proves that while surface-level grammar changes across English, Spanish, or French, the underlying "computational thought" remains mathematically consistent across languages.

2. Why Is It Timely?

2.a) *The "Black Box" Problem in Generative AI:* Current Large Language Models (LLMs) operate on heavy statistical probabilities, frequently resulting in "hallucinations" and unexplainable outputs. This work arrives at a pivotal moment when the AI industry is actively searching for Explainable AI (XAI) structures grounded in logical, symbolic systems.

2.b) *The Symbiosis of Linguistics and AI:* As AI engineering reaches the limits of pure data scaling, this paper offers a timely methodological bridge between neurolinguistics and software engineering, injecting formal linguistic theory back into modern computational models.

3. The Difference It Might Make (Broader Impact)

3.a) *Paving the Way for Explainable NLP:* By encoding concepts into structured mathematical units like conemes and axemes, this research provides a framework for AI systems that can explicitly explain their logical reasoning.

3.b) *Revolutionizing Machine Translation:* Traditional translation often breaks down due to structural discrepancies between languages. By stripping away superficial grammar to extract the core mathematical equation, this model allows for highly accurate, meaning-based vectorization that can significantly improve cross-linguistic processing.

C) Perspectives

1. Short-Term Perspectives: Algorithmic Automation. Developing an automated rule-based parser that can automatically take raw text and output its vectorized ternary and binary structures.

2. Mid-Term Perspectives: Enhancing Explainable AI (XAI). Hybrid Neuro-Symbolic AI: Modern Large Language Models (LLMs) are powerful but operate as "black boxes." A major mid-term perspective is integrating this mathematical framework into neural networks to create a hybrid system. The Semiotax equations can serve as a logical, symbolic layer that constraints the statistical outputs of LLMs, drastically reducing hallucinations.

3. Long-Term Perspectives: Cognitive and Cognitive-Neuroscience Modeling. Artificial General Intelligence (AGI) Blueprints: In the long run, formalizing "computational thought" through structured units (like 'axemes' and 'subaxemes') offers a vital methodological blueprint for building AI architectures capable of symbolic reasoning, abstract logic, and genuine conceptual understanding.